

AI Musafir International

Umrah Travel ERP — Complete Project Map

This document is a complete reference guide for the AI Musafir International travel management platform. It covers every module, every user role, how data flows between systems, key operational workflows, and a developer cost estimate — all in plain language so anyone can understand the system at a glance.

WHAT'S INSIDE

1. System Architecture Overview — how the 3 apps talk to each other
2. User Roles & Permissions — who can do what
3. ERP Module Directory — every screen, grouped by function
4. Customer Journey Map — from visitor to download
5. API Reference Map — which endpoint powers which screen
6. Database Table Directory — what data is stored and where
7. Key Operational Workflows — step-by-step process guides
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9. Estimated Development Cost — hours & cost per module in USD & PKR

BUILT WITH:

Node.js 24

TypeScript

React + Vite

Express 5

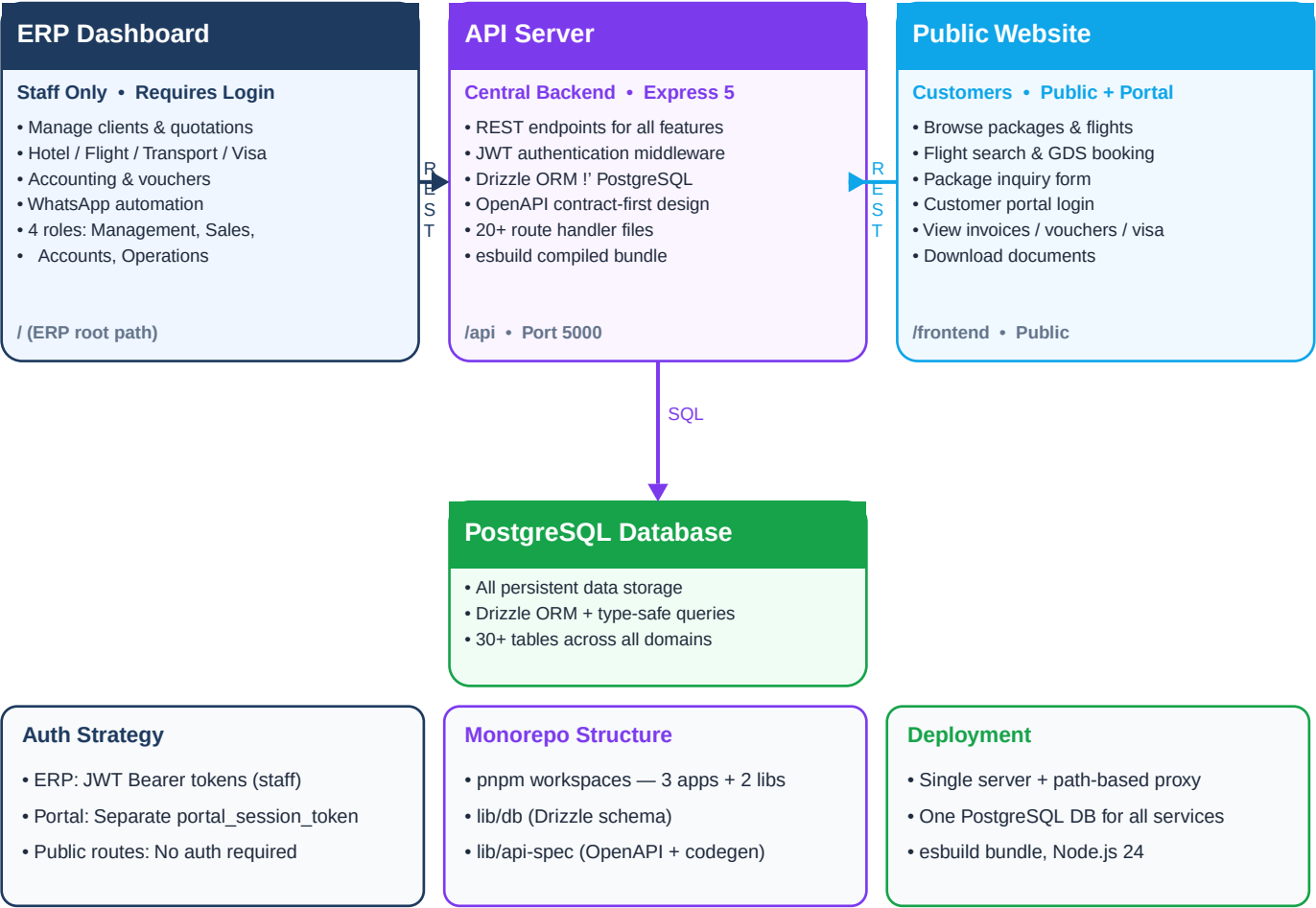
PostgreSQL

Drizzle ORM

pnpm Workspaces

System Architecture

How the three applications connect to each other and to the database



User Roles & Permissions

Who uses the system and what each role can access

Role	Who	What They Can Access
Management	Owner / Senior Management	Full access to everything — all modules, all reports, all settings, can create/edit/delete any record
Sales	Sales Executives	Clients, quotations, hotel requests, follow-ups. Cannot access accounting vouchers or system settings
Accounts	Finance / Accounts Team	Full accounting access: vouchers, ledger, P&L, balance sheet, invoices. Cannot change system settings
Operations	Operations Staff	Hotels, flights, transport, visa, bookings. View-only accounting. Cannot change system settings
Portal Customer	End Customers (self-registered)	Customer portal only: own bookings, invoices, vouchers, visa docs, transport, downloads. No ERP access
Portal DC / Agent	Distributor / Sub-agent	Portal DC login: own client bookings and documents. Separate from end-customer portal. No ERP access

HOW ROLES ARE ASSIGNED

ERP staff roles are set by a Management user in the Users section of the ERP (/users). A new staff member receives a temporary password and must change it on first login. Portal customers self-register at /portal-register on the public website — their account starts as 'Pending Approval' and a staff member must activate it from /portal-users in the ERP before they can log in.

SECURITY FEATURES

Forced Password Change	New staff accounts are flagged 'must change password'. The server blocks all API calls (except /change-password) until the password is changed.
Bcrypt Password Hashing	All passwords are stored as bcrypt hashes (\$2b\$). Plain-text passwords from old accounts are automatically re-hashed on server startup.
Login Audit Log	Every login attempt (success or failure), password change, and reset is written to the auth_audit_log table with timestamp and IP.
Separate Portal Auth	Portal customers use a completely separate authentication system (portal_session_token) — a portal token cannot access ERP endpoints.
Route-Level Guards	Every ERP API endpoint requires requireAuth middleware. Public routes (packages, currency rates, website config) are explicitly opted-out.

ERP Module Directory

Every screen in the staff dashboard, grouped by function

CRM & Sales

Dashboard	/	Summary stats: today's invoices, pending quotations, alerts
Clients	/crm	Master list of all clients/parties — add, edit, view profile
Client Profile	/crm/:id	Full client detail: bookings, invoices, documents, history
Follow-Ups	/crm/follow-ups	Pending follow-up reminders for the sales team
Quotations	/quotations	Create & manage price quotations; PDF generation
Pending Quotes	/quotations/pending	Quotations awaiting client confirmation
Hotel Requests	/hotel-requests	Incoming hotel availability requests from clients

Operations

Hotels Directory	/hotels	Hotel master list: name, city, stars, rooms, WhatsApp
Vendors	/vendors	Vendor/supplier master list
Transport	/transport	Transport bookings: vehicle type, vendor, cost, status
Flights	/flights	Flight records: sector, airline, PNR, pax, fares
Flight Cancellations	/flights/cancellations	Cancelled flight tracker
BSP Report	/flights/bsp-report	IATA BSP billing summary report
Passenger List	/flights/passengers	All passenger records across flights
Visa Applications	/visa	Visa tracking: type, applicant, status, documents

Finance & Accounting

Invoice List	/accounting/invoices	All hotel DN invoices — list, filter, search
Hotel Invoice Form	/accounting/hotel-invoice/:id	Create/edit hotel invoice; 3 print formats
Vouchers	/accounting/vouchers	Double-entry vouchers: RV/PV/JV/CV types
General Ledger	/accounting/ledger	Account-level transaction ledger
Trial Balance	/accounting/trial-balance	Debit/credit balances for all accounts
P&L Statement	/accounting/pnl	Profit & Loss report by date range
Balance Sheet	/accounting/balance-sheet	Assets, liabilities, equity snapshot
Financial Years	/financial-years	Open/close accounting periods
Currency Settings	/currency-settings	Exchange rates, multi-currency config

WhatsApp & Communications

WhatsApp Inbox	/whatsapp-inbox	Read incoming WhatsApp messages
Bot Campaigns	/bot-campaign	Automated message campaigns to groups/contacts
Media Library	/media-library	Uploaded images/docs for WhatsApp sends

Administration & Settings

Users	/users	ERP staff accounts: add, role, reset password
Portal Users	/portal-users	Customer portal accounts: approve, view, manage
Website Settings	/website-settings	Branding, WhatsApp buttons, homepage config
ERP Settings	/erp-settings	ERP-level configuration
Automation Settings	/automation-settings	WhatsApp automation rules
GDS Settings	/gds-settings	Flight GDS API credentials
AI Settings	/ai-settings	OpenAI API key for OCR/AI features

Customer Journey Map

The end-to-end path a customer takes from first visit to document download



API !” Module Map

Which API route file powers which part of the system

API Route File	ERP / Website Page	What it Does
auth.ts	ERP Login, /users (change/reset password)	JWT login, logout, me, change-password, reset-password
users.ts	ERP /users	Staff user CRUD, role management
portal.ts + portal-customer.ts	Portal /portal-login, /portal/dashboard	Portal auth, dashboard, bookings, invoices, docs
portal-users.ts	ERP /portal-users	Approve/reject/list portal accounts
clients.ts	ERP /crm, /crm/:id	Client/party CRUD, search, profile data
quotations.ts	ERP /quotations, /quotations/pending	Quotation create/edit/status/PDF
booking-inquiries.ts	ERP, Public website inquiry form	Store & list flight/booking inquiries
package-inquiries.ts	Public website package inquiry	Store & list package inquiry submissions
hotel-invoices.ts	ERP /accounting/invoices, /hotel-invoice/:id	DN invoice CRUD, next-DN generation
hotels.ts	ERP /hotels, Public /public/hotels	Hotel master CRUD, public listing
vendors.ts	ERP /vendors	Vendor/supplier CRUD
flights.ts	ERP /flights and sub-pages	Flight record CRUD, BSP, cancellations, passengers
transport.ts	ERP /transport	Transport booking CRUD
visa.ts	ERP /visa	Visa application tracking CRUD
vouchers.ts	ERP /accounting/vouchers	Double-entry voucher create/edit/post
accounting.ts	ERP /accounting/ledger, trial-balance	Chart of accounts, ledger, trial balance
accounting-reports.ts	ERP /accounting/pnl, /balance-sheet	P&L and balance sheet reports
financial-year.ts	ERP /financial-years	Financial year CRUD, open/close
currency-settings.ts	ERP /currency-settings, /api/currency/rates	Exchange rates, live rate proxy (public)
whatsapp.ts	ERP /whatsapp-inbox, /bot-campaign	WhatsApp send, session, groups, inbox
automations.ts + ai-settings.ts	ERP /automation-settings, /ai-settings	Automation rules, OpenAI key management
website-config.ts	ERP /website-settings, /api/website-config (public)	Branding, homepage config (public read)

Database Table Directory

Every table in PostgreSQL — what it stores and how it connects

Core / Authentication

users	ERP staff accounts	Linked to: vouchers, invoices, quotations (salesman)
portal_users	Customer/DC portal accounts	Linked to: clients (party)
auth_audit_log	Every login/logout/password event	Linked to: users

CRM & Sales

clients	Client/party master	Linked to: quotations, invoices, portal_users
quotations	Price quotations with status	Linked to: clients, users (salesman)
hotel_requests	Hotel availability requests	Linked to: clients, users
booking_inquiries	Inbound flight/booking inquiries	Standalone + client link
package_inquiries	Inbound package inquiries from website	Standalone

Operations

hotels	Hotel master: name, city, stars, rooms	Linked to: hotel_invoices
vendors	Supplier/vendor master	Linked to: hotel_invoices, transport_bookings
hotel_invoices	Hotel DN invoices	Linked to: clients, vendors, hotels, users
flights	Flight segment records	Linked to: clients, passengers
group_tickets	Group flight ticket records	Linked to: flights
transport_bookings	Transport bookings	Linked to: clients, vendors
visa_applications	Visa applications	Linked to: clients
passenger_documents	Uploaded passport/visa scans	Linked to: clients, portal_users

Finance & Accounting

accounts	Chart of accounts (assets/liab/equity/income/expenses)	Linked to: voucher_lines
vouchers	ERP/PPV/CV voucher headers	Linked to: accounts, users
voucher_lines	Individual debit/credit lines	Linked to: vouchers, accounts
financial_years	Accounting period definitions	Linked to: vouchers
currency_settings	Exchange rates and base currency config	Standalone

Communications & System

whatsapp_groups	WhatsApp group targets	Standalone
whatsapp_messages	Sent/received message log	Linked to: clients
bot_campaigns	Automated campaign definitions	Linked to: whatsapp_groups
automation_configs	Automation rule settings	Standalone
media_library	Uploaded media files for WhatsApp	Standalone
website_config	Key-value store for branding & settings	Standalone

Key Operational Workflows

Step-by-step process guides for the most common staff tasks

1. Create & Confirm a Quotation

- 1 Go to CRM (/crm) and open the client's profile, or create a new client.
- 2 Click 'New Quotation' !' fill in the package details, hotel, nights, pax count.
- 3 Add hotel cost (payable) and selling price (receivable) — system calculates profit.
- 4 Save as 'Draft'. Review and update if needed.
- 5 Change status to 'Sent' when shared with the client.
- 6 When client confirms, update status to 'Confirmed'. This triggers an option date alert.
- 7 Proceed to raise a Hotel Invoice (DN) from the Accounting module.

2. Raise a Hotel Invoice (DN)

- 1 Go to Accounting !' Invoices (/accounting/invoices) !' click 'New Hotel Invoice'.
- 2 Select the client (Party) and hotel vendor from the dropdowns.
- 3 Fill in hotel details: check-in/out dates (Nights auto-calculates), rooms, bed type.
- 4 In the Calculation section, enter the Rate per Night per Room (SAR).
- 5 System auto-multiplies: Rate × Nights × Rooms = Receivable Total.
- 6 Enter the PKR exchange rate or click 'live' to fetch the current rate.
- 7 Click Accept. The DN number (e.g. DN-0042) is assigned automatically.
- 8 Print the formal invoice, DN invoice, or accommodation voucher using the print buttons.

3. Post a Payment Voucher (RV / PV)

- 1 Go to Accounting !' Vouchers (/accounting/vouchers) !' click 'New Voucher'.
- 2 Select type: RV (Receipt Voucher = money received), PV (Payment Voucher = money paid).
- 3 Enter Date, Narration (description of the transaction).
- 4 Add at least 2 lines: one DEBIT line and one CREDIT line. Totals must balance (DR = CR).
- 5 Select the account for each line from the Chart of Accounts dropdown.
- 6 Click Save. The voucher is posted to the General Ledger immediately.
- 7 Verify in Ledger (/accounting/ledger) by filtering on the relevant account.

4. Send a WhatsApp Message to a Vendor

- 1 Go to WhatsApp Inbox (/whatsapp-inbox).
- 2 Ensure WhatsApp is connected (scan QR code once; session persists on server).
- 3 Click 'New Message' !' select the vendor's saved contact/group.
- 4 Type the message or select a template from Media Library.
- 5 Click Send. Message appears in the inbox log with delivery status.

5. Portal Customer: Register !' Access Documents

- 1 Customer visits the public website and clicks 'Customer Portal' or /portal-register.
- 2 Fills in name, email/phone, and password. Account created with status 'Pending Approval'.
- 3 Staff goes to ERP !' Portal Users (/portal-users) and clicks 'Approve'.
- 4 Customer receives confirmation and logs in at /portal-login.
- 5 Portal dashboard shows their bookings, invoices, vouchers, visa status, transport.
- 6 Customer downloads travel documents from /portal/downloads.

Developer Quick-Reference

Where to find or change anything in the codebase

If You Need To...	Look In / Go To	Notes
Add a new ERP page / screen	artifacts/umrah-erp/src/pages/	Create a new .tsx file, add a route in App.tsx
Add a new public website page	artifacts/frontend/src/pages/	Create .tsx file, add route in frontend App.tsx
Add or change a database table	lib/db/src/schema/	Edit the Drizzle schema file, then run: pnpm --filter @workspace/db run push
Add a new API endpoint	artifacts/api-server/src/routes/	Add route handler, register in index.ts, update OpenAPI spec
Update the OpenAPI spec	artifacts/api-spec/openapi.yaml	Edit spec, then run: pnpm --filter @workspace/api-spec run codegen
Change company branding (print)	ERP ! Website Settings (/website-settings)	Update keys in website_config table via the settings UI
Change company logo	ERP ! Website Settings ! Logo Upload	Uploads to object storage; keys: logo_url, print_logo_url
Add a WhatsApp automation rule	ERP ! Automation Settings (/automation-settings)	Configure trigger/action pairs in the UI
Change exchange rates	ERP ! Currency Settings (/currency-settings)	Manual entry or live fetch from currency API
Reset a staff password	ERP ! Users (/users) ! Reset Password	Sends a temporary password; user must change on login
Approve a portal customer	ERP ! Portal Users (/portal-users)	Click Approve on pending registration
View accounting ledger	ERP ! Accounting ! Ledger (/accounting/ledger)	Filter by account code or date range
Generate a P&L report	ERP ! Accounting ! P&L (/accounting/pl)	Select date range; exports to print
Check server logs	Workflow console / deployment logs	Use the Replit workflow console for live logs
Run a DB migration (dev)	Terminal: pnpm --filter @workspace/db run push	Pushes schema changes to the development database
Build the API server	Terminal: pnpm --filter @workspace/api-server run build	Compiles TypeScript via esbuild to dist/
Run full typecheck	Terminal: pnpm run typecheck	Typechecks all workspace packages
Regenerate API hooks & schemas	Terminal: pnpm --filter @workspace/api-spec run codegen	Regenerates React Query hooks from OpenAPI spec

PROJECT FOLDER STRUCTURE

artifacts/api-server/ Express API server — all backend code	artifacts/umrah-erp/ Staff ERP dashboard (React + Vite)
artifacts/frontend/ Public website + customer portal (React + Vite)	lib/db/ Drizzle ORM schema + migrations (shared library)
lib/api-spec/ OpenAPI contract + generated client hooks	scripts/ Utility scripts (e.g. this PDF generator)

Estimated Development Cost

Market rate estimate for a mid-senior full-stack developer team • Exchange rate: 1 USD = 290 PKR

Module / Area	Features Included	Hrs	\$/hr	Cost (USD)	Cost (PKR)
Project Setup & Infrastructure	Monorepo, API server, DB schema, CI, deployment config	40	\$35	\$1,400	Rs 406,000
Authentication & User Roles	Login, sessions, roles, forced password change, audit log	30	\$35	\$1,050	Rs 304,500
CRM / Clients	Client list, profile, follow-ups, hotel requests	35	\$35	\$1,225	Rs 355,250
Quotations	Create/edit quotation, PDF generation, status workflow, pending approvals	45	\$35	\$1,575	Rs 456,750
Hotel Invoices (DN)	Invoice form, rate×nights auto-calc, 3 print formats, status tracking	50	\$35	\$1,750	Rs 507,500
Flight Management	Flight records, BSP report, cancellations, passenger manifest	40	\$35	\$1,400	Rs 406,000
Transport Bookings	Transport invoice form, vendor assignment, status tracking	25	\$35	\$875	Rs 253,750
Visa Applications	Visa tracking, document upload, status workflow	25	\$35	\$875	Rs 253,750
Hotels & Vendors Directory	Hotel master list, vendor list, room types, WhatsApp numbers	20	\$35	\$700	Rs 203,000
Double-Entry Accounting	Chart of accounts, RV/PV/JV/CV vouchers, ledger, trial balance, P&L, balance sheet	70	\$35	\$2,450	Rs 710,500
Financial Years	Year open/close, period locking, reporting periods	15	\$35	\$525	Rs 152,250
Currency Settings	Multi-currency support, live rate fetching, PKR conversion	15	\$35	\$525	Rs 152,250
WhatsApp Automation	Inbox, bot campaigns, media library, group messaging, session persistence	60	\$35	\$2,100	Rs 609,000
Customer Portal	Portal login/register, dashboard, invoices, vouchers, visa, transport, downloads	55	\$35	\$1,925	Rs 558,250
Public Website	Landing page, package listings, flight search, GDS booking, package inquiry	50	\$35	\$1,750	Rs 507,500
Print & Branding System	Company branding config, print layouts for all document types, logo upload	20	\$35	\$700	Rs 203,000
AI / OCR Integration	Document scanning stub, AI settings page, OpenAI key management	15	\$35	\$525	Rs 152,250
ERP Settings & Admin	Website settings, automation, GDS/AI/ERP config pages	20	\$35	\$700	Rs 203,000
Portal Users Management	Staff approval workflow, portal user list, status management	15	\$35	\$525	Rs 152,250
Testing, Bug Fixes & Polish	Cross-module testing, UI polish, error handling, security hardening	50	\$35	\$1,750	Rs 507,500

TOTAL ESTIMATED HOURS:

TOTAL COST (USD @ \$35/hr):

TOTAL COST (PKR @ 290 per USD):

695 hours

USD 24,325

PKR 7,054,250

DISCLAIMER: This is a market-rate estimate only. Actual cost varies by developer location, experience level, and team size. Project management and QA overhead typically adds 20–30% to the above figures. Does not include third-party service costs (cloud hosting, WhatsApp Business API, SMS gateway, GDS/Amadeus fees, domain/SSL).